

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA43

COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. (L2)

Assessment Tool: Application-Based Questions

Weightage: 40%

QUESTIONS:

Total Marks: 30

1: Online Symposium Portal

A college is developing an online symposium registration portal. The following HTML structure and HTTP interaction is observed:

```
<form action="/register" method="POST">
  <input type="text" name="name">
  <input type="email" name="email">
  <button>Submit</button>
</form>
```

When a student submits the form, the browser sends an HTTP request to the server, and the server responds accordingly.

Questions:

1. Identify appropriate **HTML5 semantic elements** missing in the structure.
2. Modify the structure to make it **standards-compliant**.
3. Interpret the **HTTP request generated** during form submission.
4. Analyze the **HTTP response cycle** from server to browser.
5. Suggest improvements for **usability and accessibility**.

2: Cross-Browser Compatibility Issue

A company website shows inconsistent layout across browsers. The following HTML snippet is used:

```
<html>
  <head>
    <title>Company</title>
  </head>
  <body>
    <div>
      <h1>Welcome
      <p>About us
    </div>
  </body>
</html>
```

Questions:

1. Identify **improper nesting and missing closing tags**.
2. Analyze how different browsers may interpret this structure.
3. Identify **missing semantic elements** (e.g., header, section).
4. Provide a **corrected W3C-compliant HTML structure**.
5. Suggest techniques to ensure **cross-browser compatibility**.

3: Form Submission Failure

A web application fails to send form data to the server. The following configuration is observed:

```
<form action="/submit" method="GET">
  <input type="text" name="username">
  <input type="password" name="password">
  <button type="button">Login</button>
</form>
```

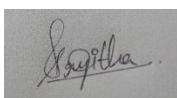
Questions:

1. Identify issues in the **form configuration**.
2. Analyze why the **HTTP request is not triggered**.
3. Interpret the role of **GET method in this scenario**.
4. Propose a **corrected implementation**.
5. Suggest improvements for **secure data transmission**.

Code of Conduct:

I s.srijitha(192471024) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

ANSWERS

QUESTION 1:

1. Online Symposium Portal

1. Identify appropriate HTML5 semantic elements missing in the structure.

Missing semantic elements include:

- <header>
- <main>
- <section>
- <article> (optional)
- <footer>
- <label> for input fields

2. Modify the structure to make it standards-compliant.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Symposium Registration</title>
</head>
<body>

<header>
  <h1>Online Symposium Registration</h1>
</header>

<main>
<section>
<form action="/register" method="POST">

<label for="name">Name:</label>
<input type="text" id="name" name="name" required>

<label for="email">Email:</label>
<input type="email" id="email" name="email" required>

<button type="submit">Submit</button>

</form>
</section>
</main>

<footer>
<p>&copy; 2026 College Symposium</p>
</footer>

</body>
</html>
```

3. Interpret the HTTP request generated during form submission.

- Browser sends an **HTTP POST** request.
- URL: /register
- Form data is sent in the **request body**.
- Example:

name=Srijitha
email=srijitha@gmail.com

4. Analyze the HTTP response cycle from server to browser.

- Browser sends POST request.
- Server receives and validates data.
- Server stores registration details.
- Server sends an HTTP response (200 OK or redirect).
- Browser displays confirmation page.

5. Suggest improvements for usability and accessibility.

- Use <label> for all input fields.
- Add required attribute.
- Use meaningful button text.
- Display validation error messages.
- Ensure keyboard accessibility.
- Use responsive design.

2. Cross-Browser Compatibility Issue

1. Identify improper nesting and missing closing tags.

Issues:

- Missing <!DOCTYPE html>
- Missing </h1>
- Missing </p>
- Missing closing </div> properly nested
- No lang attribute

2. Analyze how different browsers may interpret this structure.

- Browsers automatically close missing tags.
- Different browsers may render content differently.
- Layout inconsistencies may occur.
- Validation errors reduce compatibility.

3. Identify missing semantic elements.

Missing semantic elements:

- <header>
- <main>
- <section>
- <footer>

4. Provide a corrected W3C-compliant HTML structure.

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
<meta charset="UTF-8">
```

```
<title>Company</title>
```

```
</head>
```

```
<body>
```

```
<header>
```

```
<h1>Welcome</h1>
```

```
</header>
```

```
<main>
```

```
<section>
```

```
<p>About us</p>
```

```
</section>
```

```
</main>
```

```
<footer>
<p>Copyright 2026</p>
</footer>
```

```
</body>
</html>
```

5. Suggest techniques to ensure cross-browser compatibility.

- Use valid HTML5.
- Validate using W3C Validator.
- Use CSS Reset or Normalize.css.
- Test in Chrome, Firefox, Edge, and Safari.
- Use responsive CSS.
- Avoid deprecated HTML tags.

3. Form Submission Failure

1. Identify issues in the form configuration.

Issues:

- Button type is "button" instead of "submit".
- GET method is not secure for login.
- Missing labels.
- Missing required attributes.

2. Analyze why the HTTP request is not triggered.

- type="button" does not submit the form.
- Only type="submit" or JavaScript can submit the form.
- Therefore, no HTTP request is sent.

3. Interpret the role of GET method in this scenario.

- GET sends data in the URL.
- Username and password become visible.
- Data can be stored in browser history.
- Not suitable for login forms.

Example:

```
/submit?username=srijitha&password=12345
```

4. Propose a corrected implementation

```
<!DOCTYPE html>
<html lang="en">
```

```
<head>
<meta charset="UTF-8">
<title>Login</title>
</head>
```

```
<body>
```

```
<form action="/submit" method="POST">
```

```
<label for="username">Username</label>
<input type="text" id="username" name="username" required>
```

```
<label for="password">Password</label>
<input type="password" id="password" name="password" required>
```

```
<button type="submit">Login</button>
```

</form>

</body>

</html>

5. Suggest improvements for secure data transmission.

- Use **POST** instead of GET.
- Use **HTTPS** to encrypt data.
- Validate inputs on both client and server.
- Use strong password policies.
- Protect against SQL Injection and XSS.
- Implement authentication and session management.